

DANANG WAHYU PRASEKTIYO

AI Engineer | Machine Learning Engineer

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LinkedIn • GitHub • Portfolio • SmarTik Live Demo

PROFESSIONAL SUMMARY

AI Engineer and Computer Engineering graduate (GPA 3.78/4.00) specializing in Generative AI, LLM-based systems, and full-stack ML deployment. Built and deployed production AI applications with thoughtful system architecture, including a CycleGAN-powered batik generation platform on Google Cloud Run and a health coaching system featuring a novel anti-hallucination architecture that separates deterministic computation from LLM interpretation. BNSP-certified Associate Data Scientist with proven ability to design and ship trustworthy AI systems from model training to production deployment.

TECHNICAL SKILLS

- AI / Machine Learning:** PyTorch, TensorFlow, Scikit-Learn, LightGBM, CycleGAN, Transformers, Hugging Face, YOLO
- LLM & GenAI Stack:** LangChain, Groq API (Llama 3.3-70B), Google Gemini API, Prompt Engineering, Anti-Hallucination Architecture, Context Management
- Backend & APIs:** FastAPI, Uvicorn, RESTful API Design, Async Processing, Streamlit, Python, SQL
- MLOps & Cloud:** Google Cloud Platform (Cloud Run), Docker, Git/GitHub, CI/CD, Serverless Deployment, Supabase (PostgreSQL)
- Data & Frontend:** Pandas, NumPy, Matplotlib, Seaborn, Recharts, React.js, Next.js, Vite, TypeScript, Tailwind CSS

AI PROJECTS

SmarTik — Generative AI Platform for Batik Semarangan	2025
<i>Tech: PyTorch, CycleGAN, FastAPI, Google Cloud Run, Docker Live: smartik.id</i>	
- Led end-to-end development of a Generative AI web platform that transforms user sketches into authentic Batik Semarangan patterns, the first AI-powered cultural preservation tool for Semarang's traditional batik heritage. - Trained and fine-tuned a CycleGAN model in PyTorch on a custom-curated batik dataset, enabling unpaired image-to-image translation with high visual fidelity between sketches and generated patterns. - Architected a low-latency FastAPI backend with async preprocessing pipeline (image resizing, normalization, tensor conversion), enabling concurrent user requests without blocking. - Containerized the ML model using Docker and deployed to Google Cloud Run, achieving serverless auto-scaling and cost-efficient pay-per-request pricing. Selected as Gemastik 2025 Finalist (National ICT Competition).	
AI Health & Habit Coach — Anti-Hallucination Wellness System	2025
<i>Tech: FastAPI, LangChain, Groq API (Llama 3.3-70B), Supabase, React 18, Recharts</i>	
- Architected a novel "computation-first, LLM-second" system that eliminates hallucination on numerical health data by enforcing strict separation between deterministic Python pattern analysis and LLM language generation. - Built a pure-Python health scoring engine computing 7-14 day trend detection, per-metric scores (0-100), and weakness surfacing — feeding only pre-verified facts to the Llama 3.3-70B LLM for interpretation only. - Integrated LangChain for multi-turn conversational context (persistent 8-message history) with domain-governance prompts that ground every response in the user's actual health data profile. - Delivered a full-stack production system (FastAPI backend, React/Vite frontend, Supabase/PostgreSQL persistence) at \$0/month infrastructure cost through strategic free-tier utilization.	

WORK EXPERIENCE

Laboratory Assistant — Diponegoro University	Aug 2024 – Dec 2025
<i>Department of Computer Engineering</i>	
- Taught and mentored 150+ students per session across 5 practicum periods in 3 semesters, covering Basic Electronics, Digital Systems, and Multimedia; consistently re-appointed for technical labs based on performance. - Designed hands-on lab exercises for logic circuits and hardware troubleshooting, boosting practical comprehension and reducing debugging errors during assessments. - Managed grading workflows for 2 concurrent classes while maintaining Dean's List standing (GPA 3.78).	
Frontend Developer Intern — Diskominfo Kota Semarang	Jul 2024 – Aug 2024
<i>Semarang City Government — Communication & Informatics Office</i>	
Developed responsive UI components for internal government web applications using React.js and Tailwind CSS, ensuring cross-device compatibility across stakeholder teams.	

- Integrated RESTful APIs to connect frontend modules with backend services; directly relevant experience for building AI model demo interfaces and dashboards.

CERTIFICATIONS & TRAINING

- Associate Data Scientist — BNSP (Indonesian National Certification)**

Dec 2025 – Dec 2028

 - Certified by Indonesia's National Professional Certification Authority, validating end-to-end data science competency against national industry standards.
- Data Science Bootcamp — Digital Skola**

2024

 - Completed 3-month intensive in Python, SQL, EDA, and Machine Learning; built predictive models using Linear Regression, Classification, and K-Means Clustering on real-world datasets.

EDUCATION

- Diponegoro University — B.Eng. in Computer Engineering**

Aug 2022 – Jan 2026

 - **GPA:** 3.78 / 4.00 | **Relevant Coursework:** Artificial Intelligence, Machine Learning, Digital Systems, Data Structures & Algorithms, System Analysis & Design.
 - Honors: Gemastik 2025 (Internal Selection); PKM 2022 Internal Selection Passer.

LEADERSHIP & ORGANIZATIONS

- **Vice Chairperson — Information & Communication Unit, HIMASKOM** (Sep 2024 – May 2025): Led strategic planning and cross-stakeholder coordination across 5+ internal divisions.
- **Member — Multimedia Division, Computer Engineering Research Club** (Oct 2024 – Dec 2025): Facilitated workshops on Figma design workflows and technical mentoring.